How-To Library

Informatica

Configuring a DB2 for z/OS Resource in Metadata Manager

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Abstract

This article describes how to configure Informatica® Metadata Manager and PowerExchange® to retrieve metadata from a DB2 for z/OS system.

Supported Versions

- PowerExchange 10.4.1
- PowerCenter 10.4.1
- Metadata Manager 10.4.1

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Overview

Informatica Metadata Manager is a web-based metadata management tool that you can use to browse and analyze metadata from disparate metadata repositories. Metadata Manager helps you understand and manage how information and processes are derived. It also helps you understand the fundamental relationships between information and processes and how they are used.

Metadata Manager extracts metadata from application, business glossary, business intelligence, data integration, data modeling, and relational metadata sources. It uses PowerCenter® workflows to extract metadata from metadata sources and load it into a centralized metadata warehouse called the Metadata Manager warehouse. To extract metadata from an IBM Db2 system on z/OS, PowerCenter uses a connection to PowerExchange to retrieve the metadata.

Configuring DB2 for z/OS

Before you create a DB2 for z/OS resource, you must configure the metadata source. If you do not correctly configure the metadata source, the metadata load can fail or the metadata can be incorrectly loaded into the Metadata Manager warehouse.

To access metadata in a DB2 for z/OS subsystem, complete the following tasks:

1. Install PowerExchange.
2. Configure PowerExchange.

3. Configure DB2 for z/OS privileges.
4. Verify the DB2 for z/OS migration mode.

Install PowerExchange

Because Metadata Manager uses PowerExchange to extract metadata from a DB2 for z/OS subsystem, you must install PowerExchange.

The PowerExchange version must be the same as the Metadata Manager version. If the versions differ, the resource load can fail.

Install PowerExchange in the following locations:

- The machine where the Metadata Manager Service runs
- The z/OS system that hosts the DB2 for z/OS database subsystem runs

For more information, see the *PowerExchange Installation and Upgrade Guide*.

Configure PowerExchange

To configure PowerExchange, complete the following tasks:

- Configure PowerExchange to connect to the z/OS machine where the Metadata Manager Service runs.
- Configure the PowerExchange DBMOVER configuration file on the machine where the Metadata Manager Service runs. Add a NODE statement that points to the PowerExchange Listener on z/OS. For example:

```
/*********************************************************************/
/*  PowerExchange Configuration File
/*********************************************************************/
LISTENER=(node1,TCPIP,2480)
LISTENER=(BGQALST,TCPIP,30080)
LISTENER=(BGQALST_new,TCPIP,448)
LISTENER=(DB2zOS_SPEDDUR,TCPIP,448)
/*********************************************************************/
NODE=(DB2zOS_SPEDDUR,TCPIP,mhz890-2,3156)
NODE=(local,TCPIP,000.0.0.1,2480)
NODE=(node1,TCPIP,000.0.0.1,2480)
NODE=(BGQALST,TCPIP,mhz890-2,30080)
```

- Configure the DBMOVER configuration member on the z/OS system to control access to the DB2 for z/OS system. Add the following SECURITY statement with option 2 to use an MVS user ID and password or PowerExchange passphrase, along with your z/OS security product:

```
SECURITY=(2,N)
```

Optionally, configure the PowerExchange Listener on z/OS to recognize logins that are known to the operating system. More specifically, option 2 causes PowerExchange to control access based on the following:

- MVS user ID and password or a valid PowerExchange passphrase
- Access control features of your z/OS security product, such as RACF or ACF2

By default, PowerExchange on z/OS uses the user ID under which the PowerExchange Listener runs.

Configure DB2 for z/OS Privileges

When you load a DB2 for z/OS resource, Metadata Manager extracts metadata from the DB2 catalog tables in the DB2 for z/OS subsystem. Configure SELECT privileges on the following catalog tables in the DB2 for z/OS user account that you use to connect to the metadata source:

DB2 Catalog Tables	Metadata Extracted
SYSIBM.SYSSYNONYMS	Schemas Synonyms
SYSIBM.SYSTABLES	Schemas Tables Views Alias Columns Synonyms
SYSIBM.SYSROUTINES	Schemas Stored procedures Functions
SYSIBM.SYSSEQUENCES	Schemas Sequences Identity
SYSIBM.SYSDATATYPES	Schemas User-defined types
SYSIBM.SYSTABLESPACE	Tables Views Alias Columns
SYSIBM.SYSVIEWS	Tables Views Alias
SYSIBM.SYSCOLUMNS	Columns
SYSIBM.SYSKEYS	Primary keys Indexes
SYSIBM.SYSINDEXES	Primary keys Foreign keys Indexes
SYSIBM.SYSCHECKS SYSIBM.SYSFOREIGNKEYS	Foreign keys
SYSIBM.SYSCHECKS SYSIBM.SYSCHECKDEP	Check constraints

DB2 Catalog Tables	Metadata Extracted
SYSIBM.SYSTRIGGERS	Triggers
SYSIBM.SYSSEQUENCES	Sequences
SYSIBM.SYSSEQUENCESDEP	Identity
SYSIBM.SYSDATATYPES	User-defined types

Note: If the PowerExchange Listener does not use DB2 security, configure SELECT privileges for the PowerExchange Listener user ID instead of the DB2 for z/OS user account.

Verify the DB2 Migration Mode

Run the DB2 for z/OS database in the new-function mode. You cannot extract metadata if DB2 for z/OS runs in the compatibility mode or conversion mode.

Creating a DB2 for z/OS Resource in Metadata Manager

Use the Metadata Manager interface to create a DB2 for z/OS resource.

For more information about using the Metadata Manager interface, see the *Metadata Manager Adminiistration Guide*.

Creating an IBM DB2 for z/OS Resource

Before you create an IBM DB2 for z/OS resource, configure the IBM DB2 for z/OS source.

1. On the **Load** tab, click **Actions** > **New Resource**.

   The **Resource Selection** window appears.
2. Click **Database Management** > **DB2 zOS**.
3. Click **Next**.

   The **Properties** window appears.
4. Enter the following information:

Property	Description
Name	Name for the resource. The resource appears in the **Resources** panel with this name. The name must be unique and have from one through 256 characters. It cannot include the following characters: `/ \ : * ' ? " < > | [ ]`
Description	Description for the resource. Description cannot exceed 4000 characters.
Hide in Summary Lineage	Hides the resource in the summary view of data lineage diagrams. Enable this option to hide the resource and its child objects in the summary view. Disable this option to display the resource and its child objects in the summary view. Default is disabled.

5. Click **Next**.

The **Configuration** window appears.

6. Enter the configuration properties.
7. Click **Test Connection** to test the connection to the source system.

   If Metadata Manager cannot connect to the source system, an error message appears. Correct any error and test the connection again.
8. Click **Next**.

   The **Parameters** window appears.
9. To add schemas to the **Selected Schemas** list, select the schema in the **Available Schemas** list and click **Add**.
10. To add all available schemas, click **Add All**.
11. To remove schemas from the **Selected Schemas** list, select the schema in the **Selected Schemas** list and click **Remove**.
12. To remove all schemas from the **Selected Schemas** list, click **Remove All**.
13. Click **Next**.

    The **Schedules** window appears.
14. To add a schedule, select **Attach a Schedule** and select a schedule in the **Schedule** list.

    If you have not created a schedule, you can assign a schedule to a resource after you create the resource.
15. Click **Finish**.

The resource appears in the **Resources** panel on the **Load** tab.

IBM DB2 for z/OS Configuration Properties

Update the configuration properties when you create or edit an IBM DB2 for z/OS resource.

The following table describes the IBM DB2 for z/OS configuration properties:

Property	Description
User name	z/OS user ID.
Password	Password for the z/OS user ID.
Location	Node name in the dbmover.cfg file on the machine where the Metadata Manager Service runs that points to the PowerExchange Listener on the z/OS system.
Subsystem ID	Identifier for the IBM DB2 for z/OS subsystem.
Encoding	Code page for the IBM DB2 for z/OS subsystem. Only code pages that are compatible with the code page of the Metadata Manager repository are displayed.

Reference

After you create and load an IBM DB2 for z/OS resource, you can view and run data lineage on the metadata objects that Metadata Manager extracted.

Functions

Metadata Manager imports the following IBM DB2 for z/OS functions:

- Built-in
- User-defined
- Scalar
- Column
- Table

Metadata Manager does not import system-generated functions.

Object Names

Metadata Manager imports IBM DB2 for z/OS objects with names that do not exceed 255 characters. The name of the object includes the following components: *schema_name.table_name.object_name*.

Schemas and Qualifiers

In IBM DB2 for z/OS, the term “schema” represents distinct types, functions, stored procedures, and sequences. IBM uses “qualifier” to discuss tables, views, indexes, and aliases.

For Metadata Manager, “schema” represents the top level qualifier for all objects. The qualifier used for tables, views, indexes, and aliases is the CREATOR of the object.

Troubleshooting

If you receive an error about invalid connection information when Metadata Manager tries to retrieve the metadata from DB2 for z/OS, check that the PowerExchange connection is active and configured correctly.

- Verify that the PowerExchange Listener is running on the machine where Metadata Manager is installed.
- Check the NODE statement in the DBMOVER configuration file on the machine where Metadata Manager is installed. Make sure the NODE statement points to the PowerExchange Listener on the machine that hosts the DB2 for z/OS system.
- Run the following command to verify that PowerExchange can connect to the Listener on z/OS, using the *location* specified in the NODE statement:

  ```
  C:\>dtlrexe prog=ping loc=location
  ```
- Make sure that the DB2 user name and password are correctly specified.

Author

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